



A.3 Work, energy and power

Space, time and motion · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Work done by a constant force is $W = Fs \cdot \cos\theta$, where θ is the angle between force and displacement; this equals the energy transferred. Kinetic energy is $E_k = \frac{1}{2}mv^2$, gravitational potential energy near Earth's surface is $E_p = mgh$, and the work-energy theorem states the net work done on an object equals its change in kinetic energy. Power is the rate of energy transfer or work done, $P = W/t = Fv$ for a force moving at constant velocity v . In an isolated system total energy is conserved, though it may be transformed between forms, with efficiency = useful energy output / total energy input.

Worked example: follow the reasoning

A 800 kg elevator is lifted at a constant $2.0 \text{ m}\cdot\text{s}^{-1}$ by a motor. Find the power output of the motor needed, ignoring friction.

At constant velocity, driving force = weight = $mg = 800 \times 9.81 = 7848 \text{ N}$.

$P = Fv = 7848 \times 2.0 = 15696 \text{ W} \approx 1.6 \times 10^4 \text{ W}$.

Answer: $P \approx 1.6 \times 10^4 \text{ W}$ (16 kW).

Quick check · multiple choice

A 2.0 kg object falls freely from rest through a height of 5.0 m. What is its kinetic energy just before hitting the ground (ignore air resistance)?

A 10 J

B 49 J

C 98 J

D 196 J

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A 60 kg skier starts from rest at the top of a slope of vertical height 40 m. The slope has friction, and the skier reaches the bottom with speed $24 \text{ m}\cdot\text{s}^{-1}$. [6]
- (a) Calculate the loss in gravitational potential energy.
- (b) Calculate the kinetic energy gained at the bottom.
- (c) Calculate the energy dissipated by friction and comment on where this energy goes.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $\Delta E_p = mgh = 60 \times 9.81 \times 40 = 23544 \text{ J} \approx 2.35 \times 10^4 \text{ J}$.

(b) $E_k = \frac{1}{2}mv^2 = \frac{1}{2}(60)(24^2) = 17280 \text{ J} \approx 1.73 \times 10^4 \text{ J}$.

(c) Energy lost to friction = $23544 - 17280 = 6264 \text{ J} \approx 6.3 \times 10^3 \text{ J}$; this energy is transferred mainly to thermal energy of the skis/snow and surrounding air (and some sound).

Quick check answer

C. $E_k = E_p \text{ lost} = mgh = 2.0 \times 9.81 \times 5.0 = 98.1 \text{ J}$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.